Zaguik: Zero-Knowledge Attestation for Privacy-Preserving LLM Governance

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***Abstract*—Regulators increasingly require evidence-backed assurances about how AI systems handle sensitive information. We present Zaguik, a privacy-preserving attestation pipeline that proves, in zero knowledge, that a large language model (LLM) response passed a personally identifiable information (PII) filter, without revealing either the response text or the filtering model. The system generates an LLM response, applies a PII detection policy, and produces a zk-SNARK using EzKL over the detector inference. The proof exposes only a public commitment (a hash) to the attested output prefix, associating the proof with the encoded portion of the delivered response via a protocol-level hash commitment while keeping its contents private. We provide a runnable reference implementation, discuss practical setup requirements and ONNX compatibility constraints, and outline how such proofs can support GDPR-aligned auditing and transparent AI governance.**

I. Introduction

Artificial intelligence (AI) systems that process personal data are increasingly subject to enforceable obligations beyond voluntary transparency pledges. Regulations such as the European Union's General Data Protection Regulation (GDPR) and emerging AI acts require evidence-backed assurances about how systems handle sensitive information, comply with safety requirements, and respect data-subject rights. For large language models (LLMs) deployed in customer-facing or regulated settings, demonstrating that outputs do not leak personally identifiable information (PII) is both a legal necessity and a trust requirement. Auditors and regulators need not only policies on paper but verifiable guarantees that those policies are actually applied in production.

Today, verifying that a provider applies a PII filter to LLM outputs typically requires the auditor to trust the provider's claims or to inspect the outputs and the filtering logic directly. If the auditor sees the raw outputs, privacy is compromised; if the provider must disclose the filtering model or its parameters, intellectual property and competitive concerns arise. This tension leaves a gap: there is no cryptographically sound way to prove that a given output was filtered for PII without revealing either the output or the internals of the filter. Zero-knowledge proofs (ZKPs) address this need by allowing one party (the prover) to convince another (the verifier) that a statement is true without revealing the underlying data or computation. Applied to AI pipelines, ZKPs can attest that a specific computation was performed (e.g., that a PII filter was applied and that its result indicated no PII) while keeping the model and the actual text private.

In this work we describe and implement Zaguik, a system that produces such a verifiable, privacy-preserving attestation for LLM outputs. The pipeline works as follows: (1) an LLM generates a response to a user prompt; (2) a PII filter (in our reference implementation, a regex-based detector) is applied to the output; (3) if the output passes the filter, a SHA256 hash of the output is computed and the output is encoded as input to a circuit; (4) a zk-SNARK proof is generated using EzKL, attesting that the certified circuit was executed on the encoded output prefix and that it returned a `PASS` result; (5) any verifier holding the proof and the published hash commitment can confirm correct circuit execution over the attested prefix without seeing the text or filter internals. The proof thus attests correct execution of the circuit over the encoded prefix, while the hash commitment associates the attestation with a specific output at the protocol level. This approach illustrates how cryptographic attestation can support transparent AI governance: compliance and audit can be based on verifiable proofs rather than on trust alone, and both user privacy and provider confidentiality can be preserved. We provide a runnable reference implementation and discuss practical considerations.

II. Related Work

A. Zero-Knowledge Proof Systems

Zero-knowledge proof systems such as Groth16 [**?**], PLONK [**?**], and related zk-SNARK constructions provide succinct and efficiently verifiable proofs of correct computation over arithmetic circuits. These systems allow a prover to convince a verifier that a computation was executed correctly without revealing private inputs.

More recently, practical toolchains such as libsnark [**?**], Halo2 [**?**], and EzKL [**?**] have enabled developers to compile high-level programs and machine learning models into circuits suitable for zk-SNARK proving.

However, these works focus on the cryptographic machinery for proving circuit execution correctness. They do not address the governance question of how such proofs can be used to attest enforcement of content filtering policies over LLM outputs without revealing those outputs.

B. Zero-Knowledge Machine Learning (ZKML)

Zero-knowledge machine learning focuses on proving correct execution of machine learning inference without revealing private inputs. zkCNN [**?**] demonstrates how convolutional neural network predictions and accuracy can be verified in zero knowledge by expressing inference as arithmetic circuits.

Similarly, vCNN [?] presents a framework for generating zk-SNARK proofs that certify correct execution of convolutional neural network inference.

These works address inference correctness: proving that a model's output was computed faithfully according to fixed parameters. In contrast, our work does not aim to prove correct execution of the LLM itself. Instead, we focus on attesting enforcement of a post-generation filtering policy over LLM outputs. To our knowledge, prior ZKML research does not address governance-level attestation of output filtering for generative models, where the goal is not to prove the model computation but to prove that a compliance constraint was applied to the generated output.

C. Privacy-Preserving Machine Learning and Inference

Research in privacy-preserving machine learning includes techniques such as secure multiparty computation [?], homomorphic encryption for neural network inference [?], [?], and confidential computing enclaves.

These approaches focus on protecting either model parameters or user inputs during inference. For example, homomorphic encryption allows inference over encrypted data without revealing the input, while secure multiparty computation distributes trust across multiple parties.

Our work differs in focus: rather than protecting model inputs or parameters, we aim to protect the model output while still enabling verifiable policy enforcement. The output remains private, yet a verifier can be assured that a specified constraint was checked.

D. AI Auditing, Transparency, and Governance

AI governance frameworks emphasize auditing, transparency, and accountability. Model cards [?], datasheets for datasets [?], and regulatory frameworks such as the EU AI Act propose documentation-based or process-based accountability.

In practice, auditing of deployed systems typically requires access to logs, outputs, or internal documentation. Such approaches rely on institutional trust and do not provide cryptographic guarantees of enforcement at runtime.

Recent discussions in AI safety research have proposed verifiable AI or cryptographic attestation mechanisms, but concrete implementations for generative output filtering remain limited.

Our work contributes a concrete mechanism for cryptographically attesting that a filtering policy was executed on a specific LLM output, without revealing that output to the verifier.

E. PII Detection and Content Moderation

Personal data detection is commonly performed using rule-based systems (regex patterns), named entity recognition models, or hybrid approaches [?]. Production systems for content moderation rely on deterministic filters, ML classifiers, and policy enforcement pipelines.

However, these systems provide enforcement but not verifiability. An external auditor must either trust the system operator or inspect outputs directly. To our knowledge, prior work on PII filtering does not provide a mechanism for cryptographically proving that filtering was applied to a given output without revealing the output itself.

F. Positioning Statement

To our knowledge, no prior work combines zero-knowledge proofs with LLM output filtering to provide privacy-preserving attestation of policy enforcement. Existing zk-SNARK and ZKML systems enable proving correct circuit execution, but they do not address governance-level questions of output compliance. Conversely, AI auditing and moderation systems enforce policies but lack cryptographic verifiability.

Our work occupies the intersection of these domains: we demonstrate a practical prototype that attests enforcement of an output constraint over LLM-generated text while preserving output confidentiality.

III. Methodology

The core contribution of this work is the *cryptographic attestation pipeline*: we show how to produce a zk-SNARK proof that a given string was passed through a certified check, with the proof bound to a commitment to the output and with no disclosure of the text or the checker's internals. The PII filter is the policy we attest to; its design is deliberately simple so that the focus remains on the proof system. Below we describe the architecture, the filter used in the current implementation, the encoding of outputs for the circuit, the proof system and visibility semantics, and the setup, proof, and verification procedures.

A. System Overview

The pipeline has two phases: a *filtering phase* and a *proof phase*. In the filtering phase, a user prompt is answered by an LLM (in our implementation we use DistilGPT-2); the response is then checked by a PII detector. Only if the detector reports that the text contains no PII do we enter the proof phase. There we encode the (clean) output as input to a circuit that models the same filtering logic, generate a zk-SNARK proof over that circuit using EzKL, and output a proof published together with a public commitment (hash) of the attested output prefix. A verifier receiving the proof and the commitment can check that the certified circuit was correctly executed on the encoded prefix, without learning the text or the filter parameters. The LLM itself is out of scope of the proof: we attest only that a given string was passed through the certified PII filter and passed. The end-to-end workflow is illustrated in Fig. ??.

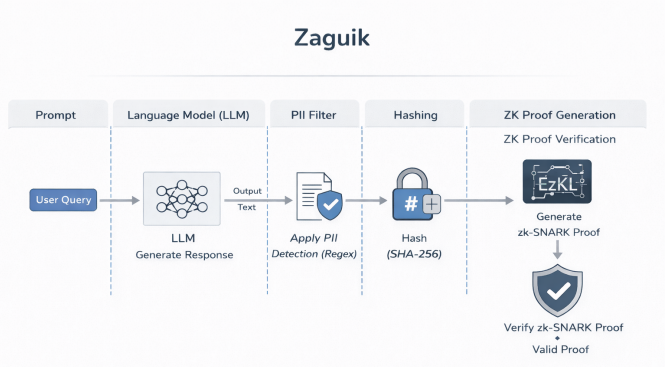

Figure 1. Zaguik's end-to-end workflow: from user prompt and LLM response through PII filtering to proof generation and verification.

B. zk-SNARK Preliminaries

A zk-SNARK (zero-knowledge succinct non-interactive argument of knowledge) is a proof system for NP relations of the form $R(x, w)$, where x is public input and w is a private witness. The prover convinces the verifier that there exists a witness w such that $R(x, w) = 1$, without revealing w. A zk-SNARK consists of three algorithms: *Setup*, which generates public parameters and keys for a fixed circuit; *Prove*, which produces a proof given the public input and private witness; and *Verify*, which checks the proof using only public data.

In our setting, the relation corresponds to correct execution of the certified circuit C on a private encoded input. The public instance consists of the circuit output (indicating pass/fail) together with protocol-level commitments. The private witness consists of the encoded output prefix and internal intermediate values required for circuit execution. The proof establishes that there exists a witness such that the circuit output equals "pass", without revealing the encoded text itself.

C. Formal Statement and Security Model

We formally define what is proven by the zk-SNARK and clarify the distinction between circuit-level guarantees and protocol-level commitments.

a) Notation: Let $y \in \{0,1\}^*$ denote the full output string generated by the LLM. Let L be a fixed encoding length in bytes. Let y_L denote the prefix of y consisting of its first L UTF-8 bytes (or the full string if $|y| \leq L$). Let $\mathrm{Enc}_L(y_L)$ denote the byte-level encoding of y_L, truncated or zero-padded to length L. Let $H(\cdot)$ be a cryptographic hash function (SHA256 in our implementation). Let C be the certified arithmetic circuit implementing the circuit-level filtering policy.

b) Public Instance and Witness: The zk-SNARK proof distinguishes between public data and private witness.

- **Public instance:**
  - The public circuit output $o \in \{0,1\}$ indicating `PASS` ($o = 1$) or `FAIL`.
- **Private witness:**
  - The encoded prefix $x = \mathrm{Enc}_L(y_L)$.
  - The internal witness values required to execute circuit C.

The verification key and structured reference string are public system parameters provided to the verifier but are not part of the public instance of a specific proof.

c) Circuit Statement: The zk-SNARK proves the following NP statement:

$$\exists x \text{ such that } C(x) = 1.$$

That is, the prover demonstrates knowledge of an input x for which the certified circuit C evaluates to `PASS`. The circuit itself operates only on the encoded prefix x and does not receive the full string y.

d) Protocol-Level Commitment: Separately from the circuit, the prover computes a public commitment

$$h = H(y_L),$$

which binds the attestation to the encoded prefix y_L. The hash value h is published alongside the proof so that external parties can associate the attestation with a specific output prefix.

Importantly, the equality $H(y_L) = h$ is not verified inside the arithmetic circuit. The binding between the public commitment h and the private witness x relies on the prover computing both from the same y_L prior to proof generation. Thus, the SNARK guarantees correct execution of $C(x)$, while the hash commitment operates at the protocol level.

e) Security Model: Under the standard soundness guarantees of zk-SNARKs, no probabilistic polynomial-time adversary can produce a valid proof for $C(x) = 1$ without knowing a witness x satisfying that relation. Zero-knowledge ensures that no information about x beyond the truth of the statement is revealed.

However, because the hash commitment is not enforced inside the circuit, the binding between h and x holds under an honest-prover assumption. A malicious prover could, in principle, produce a proof for some x while publishing an unrelated hash value h. Strengthening this binding would require verifying the hash inside the circuit or using a SNARK-friendly hash function within C.

We assume a standard adversarial model in which the prover may be malicious and the verifier is honest and follows the verification algorithm. Under the soundness property of zk-SNARKs, no probabilistic polynomial-time adversary can produce a valid proof for $C(x) = 1$ without knowledge of a witness x satisfying that relation. Zero-knowledge guarantees that the proof reveals no information about x beyond the validity of the statement.

The binding between the public commitment h and the private witness x is enforced at the protocol level and therefore holds under an honest-prover assumption. A malicious prover could, in principle, generate a valid proof for some x while publishing an unrelated hash value h. Preventing this would require incorporating hash verification inside the arithmetic circuit.

f) Scope of the Guarantee.: The proof guarantees correct execution of the certified circuit C on an encoded prefix x. If $|y| \leq L$, the entire output is attested; otherwise, only the first L bytes are covered by the guarantee. The system does not prove semantic adequacy of the filtering policy, nor does it make claims about the internal behavior of the LLM. The guarantee is limited to enforcement integrity of the circuit-level constraint over the committed prefix.

D. PII Filtering and Policy

We distinguish clearly between two components of the filtering pipeline: an operational PII detector used in plain execution, and a circuit-friendly policy used for zero-knowledge attestation.

First, the system applies a regex-based PII filter in plain execution. This detector searches for structured patterns such as email addresses, phone numbers, identification numbers, and other commonly formatted personal data. Its role is operational: it acts as a gating mechanism, preventing obviously unsafe outputs from proceeding to the proof generation stage.

Second, for zero-knowledge attestation, we implement a separate, circuit-friendly policy as a small PyTorch module that is exported to ONNX and compiled into an arithmetic circuit using EzKL. This module operates at the byte level.

It takes as input a fixed-length normalized byte vector and counts occurrences of selected "suspicious" byte values (e.g., '@'). The circuit outputs a scalar count, and the policy is considered satisfied if this count is zero.

The zk-SNARK proof certifies correct execution of this ONNX-derived byte-level circuit and the fact that its output equals zero. It does not certify direct execution of the plain regex filter. The regex detector and the byte-counter circuit are separate mechanisms: the former provides practical PII detection in deployment, while the latter encodes a structurally checkable constraint that can be efficiently represented and verified within an arithmetic circuit.

It is therefore important to clarify the scope of equivalence. The proof guarantees only that the ONNX circuit was executed correctly on the encoded output. The correspondence between the plain regex-based filter and the byte-level circuit is a design choice, not a formally verified equivalence. The byte-counter policy can be interpreted as a conservative structural proxy for certain classes of PII indicators, but it does not capture the full semantic expressiveness of the regex detector. Establishing formal semantic equivalence between high-level filtering logic and its circuit representation remains an open research problem and is outside the scope of this prototype.

The attestation mechanism itself is policy-agnostic: any deterministic filtering rule that can be expressed as an ONNX model compatible with the proof backend can be certified in the same manner. In this work, the simplified byte-level policy serves to demonstrate the feasibility of zero-knowledge attestation for LLM output governance under current practical constraints.

E. Encoding and Circuit Input

The LLM output is a string. To feed it into the circuit we encode it in fixed-size form. We fix an upper bound L on the number of UTF-8 bytes (in our implementation L = 2048). The LLM generation process is configured so that attested outputs do not exceed L bytes. Therefore, for every attested output, the circuit receives the entire string.

We use a byte-level encoding: the UTF-8 representation of the string is zero-padded to length L if shorter. Each byte is normalized to [0,1] by dividing by 255. The resulting vector has shape [1, L] and is used as the single input tensor to the ONNX circuit.

The prefix y_L that passed the plain PII check is the portion encoded and fed into the circuit. If $|y| \leq L$, the entire output is attested; otherwise, only the first L bytes are covered. The proof is therefore tied to the encoded prefix rather than necessarily the full output.

Separately, we compute a SHA256 hash of the complete output string. This hash serves as a protocol-level identifier that associates the proof with a specific delivered response, under the assumption that the prover computes both from the same output. The hash is not used as circuit input; instead, it is provided as public data alongside the proof so that external parties can associate the attestation with a specific output.

F. Proof System and Visibility Semantics

We use EzKL to compile the ONNX model into an arithmetic circuit and to produce zk-SNARK proofs (succinct, non-interactive arguments of knowledge). The security and privacy guarantees depend on how inputs, outputs, and parameters are exposed.

The encoded *input* is kept private as witness data. The verifier learns only the zk-SNARK proof and the public hash commitment: the prover commits to the encoded output string, and the verifier learns this commitment but not the underlying bytes, so the LLM response stays private while the attestation is associated with a specific output via a protocol-level commitment. The *output* of the circuit is made available to the verifier so they can check that it indicates "pass" (e.g., zero PII score). The filter *parameters* are fixed in the circuit and not revealed; the verifier only learns that the same certified circuit was executed. Together, these choices ensure that we prove that the certified circuit was executed on a private encoded prefix and returned `PASS`, while only a hash commitment to that prefix is revealed.

G. Setup and Proof Generation

Setup is performed once per circuit (e.g., per ONNX model and settings). The ONNX is used to generate the proof-system settings and to compile an arithmetic circuit; a KZG structured reference string (SRS) is required and can be obtained from a trusted source or generated locally for testing. With the compiled circuit and SRS, key generation yields a proving key and a verification key.

For each clean LLM output we wish to attest, we encode it as input data, compute the witness for the circuit, and run the prover to produce the final proof. The proof, the verification key, and (in deployment) the public commitment to the output are what the verifier needs.

H. Verification

Verification is deterministic: given the proof, the verification key, the settings, and the SRS, the verifier runs the verification algorithm and accepts if and only if the proof is valid. No secret data is required. If the verifier holds the published SHA256 hash, they can associate the proof with a specific output under the honest-prover assumption; note that this correspondence is not enforced inside the circuit.

I. Implementation and Reproducibility

Our reference implementation is in Python. The PII filter is regex-based in the current experiments. The cryptographic pipeline (regex-derived ONNX export, encoding, setup, proof generation, and verification) is implemented as a single scriptable workflow. The default configuration builds the small regex-derived ONNX, runs setup, and then generates and verifies proofs for a given text string. All steps are documented so that the pipeline can be reproduced and extended; integrating a more expressive PII model would require an ONNX export that is compatible with the proof system backend.

J. Experimental Setup

All experiments were conducted locally on a single machine with the following hardware and software configuration: a Snapdragon X 12-core X1E80100 CPU at 3.40 GHz (12 cores), 16 GB of RAM, running Windows 11 (64-bit). The implementation uses Python 3.13.3 and EzKL 22.2.1. No GPU acceleration was used; all proof generation and verification steps ran entirely on CPU.

The structured reference string (SRS) was generated locally using EzKL's `gen_srs` function rather than retrieved from a remote trusted source. This choice is appropriate for a self-contained research prototype but would differ in a production deployment, where the SRS would typically be downloaded from a public trusted ceremony.

All experiments were run locally and sequentially, with no parallel execution across configurations. Each value of $L \in \{128, 256, 512, 1024, 2048\}$ was evaluated fifteen times ($n = 15$) using independently generated reference outputs from the same prompt and model. Each prompt was designed to produce a long output exceeding 2048 bytes in order to evaluate different circuit encoding sizes L. For each output, the pipeline was executed with

$$L \in \{128, 256, 512, 1024, 2048\}.$$

When $L < 2048$, only the prefix of length L bytes was encoded and attested. The mean values and sample statistics over the 15 runs are reported in Table **??**.

IV. Results

Table I reports the one-time setup costs for the circuit with $L = 2048$. This includes settings generation, circuit compilation, structured reference string (SRS) generation, witness setup, and key generation. The total setup time was 102.44 seconds. SRS generation required 77.88 seconds, and key generation required 23.77 seconds. The remaining steps each took less than one second.

Table I
One-time setup costs (regex PII ONNX, $L = 2048$).

Step	Time (s)
Settings generation	0.14
Circuit compilation	0.02
SRS generation[a]	77.88
Witness (setup)	0.63
Key generation	23.77
Total	102.44

[a]Generated locally for testing; production deployments would download the SRS from a trusted source.

Table II presents proof size, proof generation time, witness computation time, and verification time for different values of L.

Proof size remains within a narrow range across all tested values of L, varying between 21 871 bytes ($L = 256$) and 21 929 bytes ($L = 2048$) in mean. No monotonic trend is observed. The minor variations are attributable to internal serialization differences and metadata encoding in the proof object, rather than structural changes in the underlying circuit.

Proof generation time exhibits high variance across all L values, with standard deviations ranging from $\pm 6\,573$ ms ($L = 1024$) to $\pm 9\,923$ ms ($L = 256$). Mean prove times range from approximately 14 840 ms ($L = 1024$) to 19 240 ms ($L = 256$). The measurements do not show a strictly monotonic pattern with respect to L, indicating that scheduling and memory effects dominate over circuit-size effects at this scale.

Witness computation time increases monotonically with L. The mean witness time is 54.31 ms for $L = 128$, 89.36 ms for $L = 256$, 150.55 ms for $L = 512$, 268.43 ms for $L = 1024$, and 540.82 ms for $L = 2048$.

Verification time remains broadly stable across all configurations, with means ranging from 168.73 ms ($L = 512$) to 200.85 ms ($L = 256$), though individual runs show occasional spikes reflected in the observed ranges.

V. Discussion

A. Interpretation of Results

The experimental results show that zero-knowledge attestation can be executed end-to-end using a small circuit. Proof size remains nearly constant across all tested values of the circuit encoding size L, with means ranging from 21 871 bytes ($L = 256$) to 21 929 bytes ($L = 2048$), a variation of less than 0.3%. This confirms that proof size is determined by the circuit structure rather than by the amount of attested data.

Witness generation time grows monotonically with L (from 54.31 ms at $L = 128$ to 540.82 ms at $L = 2048$), reflecting the increase in the number of input commitments required as the encoding window grows. Proof generation time is the dominant cost and shows high variance across all L values (standard deviations of 6 573–9 923 ms), indicating sensitivity to local execution conditions (e.g., OS scheduling, memory pressure) rather than a strict functional dependence on L. Verification time remains broadly stable across all configurations (168–201 ms mean), consistent with the constant-size verifier computation of KZG-based SNARKs. This asymmetry implies that the computational burden is concentrated on the prover side, while verification remains comparatively lightweight.

The experiment also validates the prefix-attestation mechanism. For $L < 2048$ bytes (the full output length), only the first L bytes are encoded and attested. All 75 proof generation and verification attempts across all 15 runs and 5 values of L succeeded, confirming that the pipeline robustly supports prefix-based attestation. This flexibility allows the attested length to be tuned to circuit cost constraints while preserving verifiability, at the cost of limiting the guarantee to the committed prefix rather than the entire output.

B. Governance and Security Implications

The proposed pipeline illustrates a shift from trust-based enforcement to cryptographically verifiable enforcement. In conventional deployments, external parties must rely on the provider's claims that safety or privacy filters are correctly applied to LLM outputs. Alternatively, verification requires disclosing either the generated output or the filtering logic, which may introduce privacy or intellectual property concerns.

Our approach enables a third party to verify that a specific, predefined circuit-based filtering policy was executed on a private encoded prefix, with the corresponding output identified via a protocol-level hash commitment under the honest-prover assumption, without revealing the output itself. The proof attests to correct execution of the circuit and to satisfaction of the encoded constraint, while the output remains private to the prover.

This mechanism supports a form of privacy-preserving auditability: verification can occur without access to the generated text. While it does not by itself ensure regulatory compliance, it provides a cryptographic primitive that may be integrated into governance frameworks where evidence of enforcement is required.

Table II
PROOF SIZE AND TIMINGS VS. CIRCUIT ENCODING SIZE L (15 INDEPENDENT RUNS; $n = 15$; REFERENCE OUTPUT = 2048 BYTES, PREFIX-TRUNCATED FOR $L < 2048$). EACH L BLOCK SHOWS MEAN, SAMPLE STANDARD DEVIATION ($\pm$STD), AND OBSERVED RANGE [MIN, MAX].

L (bytes)		Proof size (bytes)	Proof gen. (ms)	Witness (ms)	Verify (ms)
128	Mean	21896.67	15240.07	54.31	195.64
	$\pm$Std	$\pm$27.33	$\pm$8049.70	$\pm$21.53	$\pm$50.65
	[Min, Max]	[21850, 21944]	[10014.94, 33601.85]	[40.91, 128.50]	[150.74, 305.65]
256	Mean	21870.73	19240.44	89.36	200.85
	$\pm$Std	$\pm$39.03	$\pm$9922.94	$\pm$18.08	$\pm$65.80
	[Min, Max]	[21822, 21966]	[11019.33, 32606.75]	[70.93, 134.03]	[149.94, 398.31]
512	Mean	21892.67	15944.43	150.55	168.73
	$\pm$Std	$\pm$33.82	$\pm$7903.57	$\pm$16.24	$\pm$22.70
	[Min, Max]	[21835, 21954]	[10983.38, 33031.74]	[122.49, 183.49]	[140.98, 232.17]
1024	Mean	21893.07	14840.25	268.43	174.02
	$\pm$Std	$\pm$32.81	$\pm$6573.20	$\pm$20.57	$\pm$34.71
	[Min, Max]	[21841, 21955]	[12137.78, 37693.64]	[237.38, 295.59]	[138.79, 260.96]
2048	Mean	21929.27	18219.90	540.82	182.45
	$\pm$Std	$\pm$37.22	$\pm$9359.92	$\pm$69.33	$\pm$30.32
	[Min, Max]	[21867, 21991]	[12659.16, 43176.02]	[447.87, 704.70]	[152.73, 259.90]

The architecture is policy-agnostic at the circuit level. Any deterministic filtering rule that can be expressed as an ONNX model compatible with the proof system can, in principle, be attested using the same mechanism. In this work, we instantiate the approach with a byte-level structural policy derived from common PII indicators, but the framework can accommodate other circuit-representable post-processing constraints.

C. Limitations

Several limitations must be acknowledged.

First, the filtering policy attested within the zero-knowledge circuit is intentionally lightweight. The circuit implements a byte-level structural constraint designed to be efficiently representable as an arithmetic circuit. While sufficient to demonstrate feasibility, it does not reflect the complexity of modern neural PII detection systems. Attempts to integrate larger ONNX-based NER models revealed compatibility constraints within current ZKML tooling.

Second, the scope of the guarantee is bounded by the encoding length L. The circuit operates on a fixed-length byte vector, and only the first L bytes of an output are attested. When L is smaller than the full output length, the guarantee applies only to the encoded prefix. Although our experiments evaluate multiple values of L, extending coverage to arbitrarily long outputs would require increasing L, segmenting outputs into chunks with multiple proofs, or employing recursive proof composition.

Third, the system proves correct execution of the ONNX-derived circuit, not direct execution of the plain regex filter used in deployment. The regex-based detector and the circuit policy are separate components. While the circuit is designed to capture structural indicators of PII, no formal equivalence proof is provided between the plain filtering logic and its circuit representation.

Fourth, consistency between the public hash commitment and the encoded circuit input is enforced at the protocol level rather than inside the arithmetic circuit. The proof certifies correct execution of the circuit on a private input, but does not cryptographically enforce that the published hash corresponds to the exact same byte sequence used within the proof. Strengthening this binding by verifying the hash inside the circuit would increase circuit complexity and proof cost.

Fifth, the system guarantees enforcement integrity of the encoded policy, not the semantic adequacy of the policy itself. A weak or incomplete detector may be correctly executed and proven. Zero-knowledge attestation ensures that a rule was applied, but does not evaluate whether the rule sufficiently captures all relevant forms of personal data.

Finally, production PII filtering policies are inherently dynamic: filters are frequently updated to address new threat patterns, evolving regulatory requirements, and emerging data types. The static nature of the arithmetic circuit means that any modification to the filtering policy requires recompiling the circuit, regenerating the proving and verification keys, and repeating the setup phase. This operational overhead may limit the practical applicability of the approach in deployment contexts where filtering logic changes frequently.

D. Towards Production Deployment

Deploying Zaguik in a real production environment would require addressing several practical requirements beyond the current prototype. We outline the key components of a minimal viable deployment.

First, the filtering circuit would need to be replaced with a more expressive policy. The current byte-level counter serves as a proof of concept but does not reflect the complexity of real PII detection. A lightweight neural NER model or a regex-derived circuit compatible with EzKL would provide stronger guarantees, provided ONNX compatibility constraints are satisfied.

Second, the structured reference string (SRS) would need to be obtained from a public trusted ceremony, such as the Perpetual Powers of Tau, rather than generated locally. Local SRS generation, as used in this prototype, does not provide the soundness guarantees required in an adversarial setting.

Third, proof generation time would need to be reduced significantly. The current 15–19 seconds per proof on CPU is not viable for high-throughput deployments. GPU acceleration, which can reduce proving time by one to two orders of magnitude, and optimized native implementations of EzKL would bring this cost into a practical range.

Fourth, proof generation should be decoupled from the real-time request pipeline through an asynchronous archi-

tecture. Proofs would be generated in the background after each response is delivered, stored alongside a SHA256 hash commitment, and made available for audit on demand. This eliminates latency concerns for end users while preserving the audit trail for regulators.

Finally, a verifier interface would be needed to allow regulators and auditors to check proofs without requiring cryptographic expertise. A simple black-box tool accepting a proof and returning VALID or INVALID would make the system accessible to non-technical compliance stakeholders.

Taken together, these components define a realistic path from the current prototype to a deployable compliance primitive, particularly suited to regulated organizations operating their own private LLM infrastructure.

VI. Conclusions

We presented Zaguik, a privacy-preserving attestation pipeline that proves, in zero knowledge, that a circuit-encoded filtering policy was executed on an LLM output prefix and returned a `PASS` result without revealing the output itself. Using EzKL and a fixed ONNX-derived circuit, we evaluated the end-to-end proving pipeline under multiple encoding lengths. Empirical results show that proof size and verification time remain stable across tested configurations, while witness computation grows with the encoding length and proof generation remains the dominant cost.

Our prototype does not attempt to prove correct execution of the LLM itself, nor does it establish formal equivalence between deployment-time filtering logic and its circuit representation. Instead, it demonstrates that enforcement of a circuit-encoded policy over model outputs can be attested cryptographically while preserving output confidentiality.

These results suggest that zero-knowledge proofs can serve as a building block for verifiable enforcement mechanisms in AI systems. In particular, such attestations could be incorporated into workflows where language model outputs trigger downstream actions, interact with external tools, or are subject to compliance requirements. In these contexts, the ability to produce cryptographic evidence that a specific constraint was applied to a given output, without revealing that output, provides a new primitive for accountability.

Future work includes integrating more expressive filtering models within circuit constraints, strengthening the binding between public commitments and circuit inputs, scaling to longer outputs through chunking or recursive composition, and evaluating integration within broader AI governance and compliance infrastructures. An additional direction is extending the attestation mechanism to agentic workflows, where intermediate model outputs trigger external actions, enabling policy checks to be cryptographically enforced before such actions are executed.

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